THE LOGARITHMIC NUMBER OF SUBGROUPS OF THE SYMMETRIC GROUP

Abstract. Let
$$s(n) = |\{H : H \le S_n\}|.$$
The main asymptotic formula is
$$\log_2 s(n) = \frac{n^2}{16} + o(n^2), \qquad \text{equivalently} \qquad s(n) = 2^{(1/16+o(1))n^2}.$$
The lower bound is elementary and comes from the subgroup lattice of an elementary abelian 2-subgroup of rank $\lfloor n/2 \rfloor$. The matching upper bound is the deep theorem of Pyber on subgroup enumeration in symmetric groups, and is explicitly quoted as an imported theorem rather than reproved here. We also prove the exact statistical statement in the elementary abelian model: a uniformly random subgroup of $C_2^{\lfloor n/2 \rfloor} \le S_n$ has order $2^{n/4+o(n)}$. Finally, we record a caution: the leading asymptotic alone does not determine the order of a uniformly random subgroup of S_n; a dihedral-block construction gives $2^{n^2/16+O(n)}$ subgroups of order $2^{n/2+o(n)}$ when $4 \mid n$.

1. Statement of the problem and scope

All logarithms are base 2. Let
$$s(n) = |\operatorname{Sub}(S_n)|$$
denote the number of subgroups of the symmetric group on n letters. The principal theorem is as follows.

Theorem 1.1. *As $n \to \infty$,*
$$s(n) = 2^{n^2/16+o(n^2)}.$$
Equivalently,
$$\log_2 s(n) \sim \frac{n^2}{16}.$$

The proof has two logically different parts. The lower bound is elementary and is proved in full below. The upper bound is a substantial finite-permutation-group theorem of Pyber. Since the proof of Pyber's theorem is long and uses nontrivial subgroup-enumeration machinery, it is stated in this note as an imported theorem. Thus every local lemma and proposition below is proved from first principles, while the global upper bound is quoted precisely as the single external input.

The statistical question is more delicate. The elementary lower-bound model has a clean and fully proved order law: a random subspace of $\mathbb{F}_2^{\lfloor n/2 \rfloor}$ has dimension $n/4 + o(n)$, hence order $2^{n/4+o(n)}$. However, this model theorem should not be confused with a theorem about a uniformly random subgroup of all of S_n. Section 6 shows that there are also $2^{n^2/16+O(n)}$ subgroups of S_n with order $2^{n/2+o(n)}$ for $4 \mid n$. Consequently, the leading logarithmic formula by itself is too coarse to determine the global order distribution.

2. Gaussian binomial coefficients

For integers $0 \le d \le r$, let
$$\begin{bmatrix} r \\ d \end{bmatrix}_2$$
denote the number of d-dimensional linear subspaces of $\mathbb{F}_2^r$. These numbers are the Gaussian binomial coefficients over $\mathbb{F}_2$.

Lemma 2.1 (Formula for $\left[\begin{smallmatrix} r \\ d \end{smallmatrix}\right]_2$)**.** *For $0 \le d \le r$,*

$$\begin{bmatrix} r \\ d \end{bmatrix}_2 = \frac{(2^r-1)(2^r-2)\cdots(2^r-2^{d-1})}{(2^d-1)(2^d-2)\cdots(2^d-2^{d-1})}.$$

Proof. Count ordered bases of d-dimensional subspaces of $\mathbb{F}_2^r$ in two ways. An ordered linearly independent d-tuple in $\mathbb{F}_2^r$ can be chosen in

$$(2^r-1)(2^r-2)\cdots(2^r-2^{d-1})$$

ways: after i independent vectors have been chosen, their span has 2^i elements, so the next vector has 2^r-2^i choices. On the other hand, each fixed d-dimensional subspace has

$$(2^d-1)(2^d-2)\cdots(2^d-2^{d-1})$$

ordered bases, by the same argument inside $\mathbb{F}_2^d$. Dividing gives the formula. $\square$

Lemma 2.2 (Uniform Gaussian-binomial estimate)**.** *There is an absolute constant*

$$C_2 = \prod_{j=1}^{\infty}(1-2^{-j})^{-1} < \infty$$

such that, for all $0 \le d \le r$,

$$2^{d(r-d)} \le \begin{bmatrix} r \\ d \end{bmatrix}_2 \le C_2\, 2^{d(r-d)}.$$

Proof. Starting from Lemma 2.1, factor out the powers of 2:

$$\begin{bmatrix} r \\ d \end{bmatrix}_2 = 2^{d(r-d)} \prod_{i=0}^{d-1} \frac{1-2^{i-r}}{1-2^{i-d}}.$$

Since $r \ge d$, one has $2^{i-r} \le 2^{i-d}$, hence

$$1-2^{i-r} \ge 1-2^{i-d}.$$

Each factor in the product is therefore at least 1, proving the lower bound.

For the upper bound, use $1-2^{i-r} \le 1$. Then

$$\prod_{i=0}^{d-1} \frac{1-2^{i-r}}{1-2^{i-d}} \le \prod_{i=0}^{d-1}(1-2^{i-d})^{-1} = \prod_{j=1}^{d}(1-2^{-j})^{-1} \le \prod_{j=1}^{\infty}(1-2^{-j})^{-1}.$$

The infinite product converges because $\sum_{j\ge1} 2^{-j} < \infty$. This proves the estimate. $\square$

Lemma 2.3 (Number of subspaces of $\mathbb{F}_2^r$)**.** *Let*

$$G(r) = \sum_{d=0}^{r} \begin{bmatrix} r \\ d \end{bmatrix}_2.$$

Then

$$G(r) = 2^{r^2/4+O(\log r)}.$$

In particular,

$$\log_2 G(r) = \frac{r^2}{4} + O(\log r).$$

Proof. The function $d(r-d)$ is maximized at $d = r/2$, with maximum $r^2/4$. Therefore Lemma 2.2 gives

$$G(r) \le \sum_{d=0}^{r} C_2 2^{d(r-d)} \le (r+1)C_2 2^{r^2/4}.$$

For the lower bound, take $d = \lfloor r/2 \rfloor$. Then

$$G(r) \ge \begin{bmatrix} r \\ \lfloor r/2 \rfloor \end{bmatrix}_2 \ge 2^{\lfloor r/2 \rfloor \lceil r/2 \rceil} = 2^{r^2/4+O(1)}.$$

Combining the two estimates gives the claim. $\square$

Lemma 2.4 (Concentration of the dimension of a random subspace)**.** *Let U_r be chosen uniformly from all linear subspaces of $\mathbb{F}_2^r$. Then*

$$\frac{\dim U_r}{r} \to \frac{1}{2}$$

in probability. Equivalently, for every fixed $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\dim U_r - \frac{r}{2}\right| \geq \varepsilon r\right) \to 0.$$

More quantitatively,

$$\mathbb{P}\left(\left|\dim U_r - \frac{r}{2}\right| \geq \varepsilon r\right) \leq (r+1)C_2\, 2^{-\varepsilon^2 r^2 + O(1)}.$$

Proof. For every d,

$$d(r-d) = \frac{r^2}{4} - \left(d - \frac{r}{2}\right)^2.$$

Thus, if $|d - r/2| \geq \varepsilon r$, Lemma 2.2 gives

$$\begin{bmatrix} r \\ d \end{bmatrix}_2 \leq C_2 2^{r^2/4 - \varepsilon^2 r^2}.$$

There are at most $r+1$ possible dimensions d, so the number of exceptional subspaces is at most

$$(r+1)C_2 2^{r^2/4 - \varepsilon^2 r^2}.$$

By Lemma 2.3, the total number of subspaces is at least $2^{r^2/4+O(1)}$. Dividing the two bounds proves the quantitative estimate and hence the probabilistic convergence. $\square$

3. The elementary lower bound for $s(n)$

Let

$$m = \left\lfloor \frac{n}{2} \right\rfloor.$$

Inside S_n, define

$$E_m = \langle (1\,2), (3\,4), \ldots, (2m-1\ 2m) \rangle.$$

Then $E_m \cong C_2^m$, because the displayed transpositions are disjoint and independent.

Proposition 3.1 (Elementary lower bound)**.** *One has*

$$s(n) \geq 2^{n^2/16 + O(n)}.$$

In fact,

$$s(n) \geq G\left(\left\lfloor \frac{n}{2} \right\rfloor\right),$$

where $G(r)$ is the number of subspaces of $\mathbb{F}_2^r$.

Proof. The subgroup E_m is naturally an m-dimensional vector space over $\mathbb{F}_2$: multiplication in E_m corresponds to addition of exponent vectors in $\mathbb{F}_2^m$. Therefore the subgroups of E_m are exactly the $\mathbb{F}_2$-linear subspaces of $\mathbb{F}_2^m$. Distinct subspaces give distinct subgroups of S_n. Hence

$$s(n) \geq |\operatorname{Sub}(E_m)| = G(m).$$

By Lemma 2.3,

$$G(m) = 2^{m^2/4 + O(\log m)}.$$

Since $m = \lfloor n/2 \rfloor$, we have

$$\frac{m^2}{4} = \frac{n^2}{16} + O(n).$$

Thus

$$s(n) \geq 2^{n^2/16 + O(n)}.$$

$\square$

4. The upper bound and the asymptotic formula

The following theorem is the non-elementary input.

Imported theorem 4.1 (Pyber's upper bound)**.** *As $n \to \infty$,*

$$s(n) \leq 2^{n^2/16+o(n^2)}.$$

Remark 4.2. Theorem 4.1 is the difficult half of the subgroup-enumeration theorem for symmetric groups. It is not an elementary consequence of the lower-bound construction above. Its proof uses substantial information about the structure and enumeration of finite permutation groups. In this manuscript it is quoted as a black box; the purpose of the surrounding arguments is to make the reduction to this single external theorem completely explicit.

Proof of Theorem 1.1. The lower bound from Proposition 3.1 gives

$$s(n) \geq 2^{n^2/16+O(n)} = 2^{n^2/16+o(n^2)}.$$

The imported upper bound, Theorem 4.1, gives

$$s(n) \leq 2^{n^2/16+o(n^2)}.$$

Combining the two inequalities yields

$$s(n) = 2^{n^2/16+o(n^2)}.$$

Taking logarithms gives

$$\log_2 s(n) = \frac{n^2}{16} + o(n^2),$$

as claimed. ∎

5. The order law inside the elementary abelian model

The lower-bound construction not only gives many subgroups; it also gives a clean order statistic inside that model.

Theorem 5.1 (Model order theorem)**.** *Let $m = \lfloor n/2 \rfloor$, and choose a subgroup $U_n \leq E_m$ uniformly at random from all subgroups of E_m. Then*

$$\log_2 |U_n| = \frac{n}{4} + o(n)$$

in probability. Equivalently,

$$|U_n| = 2^{n/4+o(n)}$$

in probability.

Proof. Under the identification $E_m \cong \mathbb{F}_2^m$, the subgroup U_n is a uniformly random subspace of $\mathbb{F}_2^m$. Its group order is

$$|U_n| = 2^{\dim U_n},$$

so

$$\log_2 |U_n| = \dim U_n.$$

By Lemma 2.4,

$$\dim U_n = \frac{m}{2} + o(m)$$

in probability. Since $m = n/2 + O(1)$, this becomes

$$\log_2 |U_n| = \frac{n}{4} + o(n),$$

which is the claim. ∎

6. Why the global order distribution is a finer question

It is tempting to infer from Theorem 5.1 that a uniformly random subgroup of S_n itself should have order $2^{n/4+o(n)}$. That inference is not justified by the leading asymptotic formula. The following construction shows why: another natural family has the same leading $2^{n^2/16}$-scale but a different linear order scale.

Lemma 6.1 (A dihedral block group)**.** *Let D_8 be the dihedral group of order 8, acting faithfully on four points as the symmetry group of a square. Its Frattini subgroup $\Phi(D_8)$, namely the intersection of all maximal subgroups of D_8, has order 2, and*

$$D_8/\Phi(D_8) \cong C_2^2.$$

Proof. Write

$$D_8 = \langle r, s : r^4 = s^2 = 1,\ srs = r^{-1}\rangle.$$

The subgroup $\langle r^2\rangle$ is central of order 2. The quotient $D_8/\langle r^2\rangle$ is generated by the images of r and s, both of order 2, and these images commute because the commutator relation becomes trivial modulo $\langle r^2\rangle$. Thus the quotient has order 4 and is isomorphic to C_2^2.

It remains to identify $\langle r^2\rangle$ with the Frattini subgroup. Since $|D_8| = 8$, the maximal subgroups have order 4. They are exactly

$$\langle r\rangle, \qquad \langle r^2, s\rangle, \qquad \langle r^2, rs\rangle.$$

Indeed, $\langle r\rangle$ is the unique cyclic subgroup of order 4, while every subgroup of order 4 not equal to $\langle r\rangle$ contains a reflection and r^2, giving one of the two displayed Klein four subgroups according to whether the reflection is $r^{2j}s$ or $r^{2j+1}s$. Their intersection is precisely $\langle r^2\rangle$. Hence

$$\Phi(D_8) = \langle r\rangle \cap \langle r^2, s\rangle \cap \langle r^2, rs\rangle = \langle r^2\rangle.$$

The quotient by $\Phi(D_8)$ is therefore C_2^2, as proved above. □

Proposition 6.2 (Many subgroups at order scale $2^{n/2}$)**.** *Suppose $n = 4k$. Then S_n contains a family $\mathcal{F}_n$ of subgroups with*

$$|\mathcal{F}_n| = 2^{n^2/16+O(\log n)}$$

such that all but an exponentially small proportion of the members of $\mathcal{F}_n$ satisfy

$$\log_2 |H| = \frac{n}{2} + o(n).$$

Proof. Partition $\{1, \dots, n\}$ into k fixed blocks of size 4. On each block let a copy of D_8 act as in Lemma 6.1, and set

$$P_k = D_8^k \le S_{4k}.$$

Let

$$N_k = \Phi(D_8)^k \trianglelefteq P_k.$$

Then

$$N_k \cong C_2^k \qquad \text{and} \qquad P_k/N_k \cong C_2^{2k}.$$

Let

$$\pi : P_k \to P_k/N_k$$

be the quotient map. For every subspace $U \le P_k/N_k$, the inverse image

$$H_U = \pi^{-1}(U)$$

is a subgroup of P_k, hence a subgroup of S_{4k}. Distinct subspaces U give distinct subgroups H_U, since $\pi(H_U) = U$. Therefore the number of such subgroups is exactly the number of subspaces of $\mathbb{F}_2^{2k}$, namely

$$G(2k) = 2^{(2k)^2/4+O(\log k)} = 2^{k^2+O(\log k)}.$$

Since $n = 4k$, this is

$$2^{n^2/16+O(\log n)}.$$

Moreover, if $\dim U = d$, then

$$|H_U| = |N_k|\,|U| = 2^k 2^d = 2^{k+d}.$$

By Lemma 2.4, a uniformly random subspace $U \le \mathbb{F}_2^{2k}$ has

$$d = k + o(k)$$

in probability. Hence, for all but an exponentially small proportion of the subspaces U,

$$\log_2 |H_U| = k + d = 2k + o(k) = \frac{n}{2} + o(n).$$

This proves the claim. $\square$

Corollary 6.3 (The leading formula alone does not give a global order law)**.** *The asymptotic formula*

$$s(n) = 2^{n^2/16+o(n^2)}$$

does not by itself imply that a uniformly random subgroup of S_n has order $2^{n/4+o(n)}$. Indeed, for $4 \mid n$, there is a family of

$$2^{n^2/16+O(\log n)}$$

subgroups of order scale $2^{n/2+o(n)}$, as shown in Proposition 6.2.

Proof. Theorem 5.1 proves the order scale $2^{n/4+o(n)}$ only for subgroups chosen inside a fixed elementary abelian subgroup of rank $\lfloor n/2 \rfloor$. Proposition 6.2 constructs, on the same leading $2^{n^2/16}$ logarithmic scale, subgroups whose order is instead $2^{n/2+o(n)}$. Therefore the leading exponential count cannot distinguish the global probability weights of these different families. A genuine theorem for a uniformly random subgroup of S_n requires finer enumeration than the main logarithmic asymptotic. $\square$

7. A useful rank check for elementary abelian 2-subgroups

The following elementary observation explains why the rank $\lfloor n/2 \rfloor$ appearing in the lower bound is the correct maximum possible rank for elementary abelian 2-subgroups of S_n.

Lemma 7.1. *If $A \le S_n$ is elementary abelian of exponent 2, say $A \cong C_2^r$, then*

$$r \le \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. Let A act on $\{1, \dots, n\}$, and decompose the set into A-orbits

$$\Omega_1, \dots, \Omega_t.$$

For each orbit Ω_i, the image of A in $\mathrm{Sym}(\Omega_i)$ is a transitive abelian permutation group. A transitive abelian permutation group acts regularly after factoring out its kernel: if an element fixes one point of a transitive orbit, then, because the group is abelian, it fixes every point of that orbit. Hence the faithful image on Ω_i is an elementary abelian 2-group of order $|\Omega_i|$. Thus $|\Omega_i| = 2^{a_i}$ for some $a_i \ge 0$, and the rank of the image on Ω_i is a_i.

The diagonal action of A on all its orbits is faithful, so the natural homomorphism from A to the product of its images on the orbits is injective. Therefore

$$r \le \sum_{i=1}^{t} a_i.$$

For each orbit, $a_i \le 2^{a_i}/2$ when $a_i \ge 1$, and the inequality is also true for $a_i = 0$. Hence

$$r \le \sum_{i=1}^{t} a_i \le \frac{1}{2} \sum_{i=1}^{t} 2^{a_i} = \frac{n}{2}.$$

Since r is an integer, $r \le \lfloor n/2 \rfloor$. $\square$

8. Verification of the proof

We finish by isolating exactly what has been proved and where each input enters.

(1) Lemmas 2.1–2.4 are fully elementary linear algebra over $\mathbb{F}_2$. They prove both the count and dimension concentration for subspaces of $\mathbb{F}_2^r$.

(2) Proposition 3.1 injects the subgroup lattice of $C_2^{\lfloor n/2 \rfloor}$ into $\mathrm{Sub}(S_n)$. This proves the lower bound
$$s(n) \geq 2^{n^2/16+O(n)}.$$

(3) The only imported result is Theorem 4.1, Pyber's upper bound
$$s(n) \leq 2^{n^2/16+o(n^2)}.$$
Combining it with the elementary lower bound proves
$$s(n) = 2^{n^2/16+o(n^2)}.$$

(4) Theorem 5.1 is a fully proved statistical theorem for the elementary abelian lower-bound model. It says that a random subgroup of $C_2^{\lfloor n/2 \rfloor} \leq S_n$ has order $2^{n/4+o(n)}$.

(5) Proposition 6.2 proves that the global order distribution of a uniformly random subgroup of S_n cannot be read off from the leading logarithmic formula alone: for $4 \mid n$, there is also a family of $2^{n^2/16+O(\log n)}$ subgroups with order $2^{n/2+o(n)}$.

Thus the asymptotic formula for the number of subgroups is rigorous once Pyber's upper-bound theorem is admitted, while the order-distribution question is explicitly separated from the coarser logarithmic enumeration.

References

[1] L. Pyber, *The number of subgroups of the symmetric group*, Combinatorica **13** (1993), 363–372.